Task 1 : Conceptual Understanding

1. What is the difference between "Love" and "love" in NLP?

Ans. - In NLP, "Love" and "love" are treated as different words because of case sensitivity.
- "Love" (with capital L) is usually used at the beginning of a sentences or as a proper noun, while "love" is used as a normal word.
- To avoid this difference, NLP systems often convert all text into lowercase. This process is called case normalization

2. What happens if stopwords are not removed?

Ans. - Stopwords are common words such as "is", "the", "and" etc. If stopwords are not removed, the data becomes noisy and less meaningful.
- It also increases processing time and reduces the efficiency of the model because important words get less importance.

Provide Two real-world Scenarios where removing stopwords can be harmful.

ⓐ Sentiment Analysis :-
In the sentences "This image is not good", the word "not" is very important. If it is removed, the sentences becomes "This image is good", which completely changes the meaning.

ⓑ Question Answering / Search Engines :-
In the question "What is the time now?", if stopwords are removed, it becomes "time now", which loses the actual meaning and structure of the question.

Difference between Stemming and lemmatization

Stemming :- It is a Process of reducing words to their root form by removing suffixes. It is fast but may Produce incorrect or non-meaningful words.

Lemmatization :- It converts words into their correct base ~~from~~ form using a dictionary and grammar rules. It is more accurate and meaningful.

Stemming is faster but less accurate. while lemmatization is slower but gives correct and Meaningful words.